

MAKERERE UNIVERSITY
SCHOOL OF COMPUTING AND INFORMATICS
TECHNOLOGY

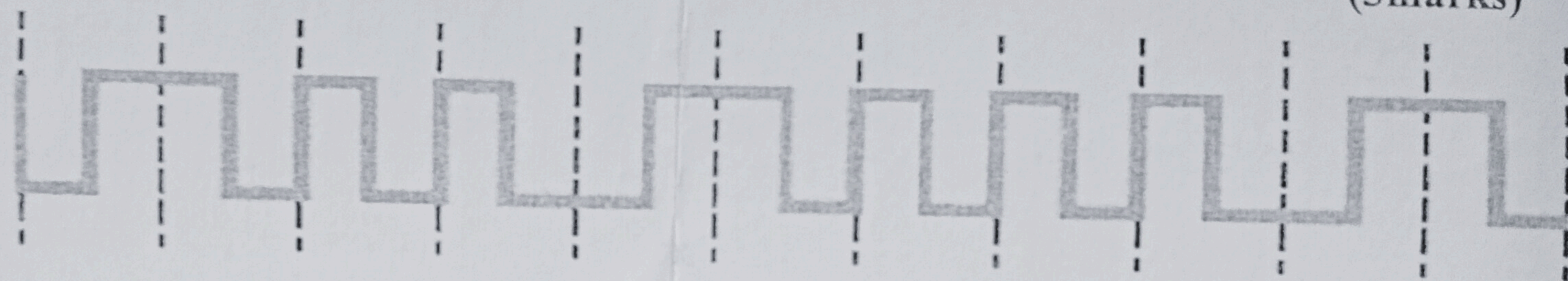
Data Communications BSE 2206

TIME: 01:00-02:00 pm 07th March 2025

INSTRUCTIONS: Attempt all questions

Time Allowed: One Hour

- a. We have a channel with 4 GHz bandwidth. If we want to send data at 20Gbps, calculate the number of signal levels, SNR and SNR_{dB}, and number of bits per level. (7marks)
- b. What is data communication and mention the four fundamental characteristics of effective data communication? (3marks)



- c. The signal in the diagram above is a Differential Manchester. Decode the signal and encode the output bits using NRZ-L, Pseudoternary, and Manchester (5marks)
- d. The attenuation of a signal is -10dB. What is the final signal power if it was originally 2W? (3marks)
- e. What are the propagation and transmission times for a 3-GB message (an e-mail) if the network bandwidth is 21 Mbps? Assume that the distance between the sender and the receiver is 90,000 km and that light travels at 3.0×10^8 m/s. (6marks)
- f. Thermal noise is one of the types of noise, and it can also be referred to as white noise. Explain why it is called white noise and mention any three other types of noise. (4marks)
- g. The period of a signal is 10ns, and the speed of light is assumed to be 3.0×10^8 m/s. Calculate the wavelength of the signal in mm (3marks)
- h. With the aid of a diagram, differentiate between the three types of transmission impairment in guided media. (6marks)
- i. Write brief notes about the following terms (1mark@)
- i. Polar
 - ii. Pulse rate
 - iii. Length of a bit